(Please write your Enrollment Number)

End-Term Examination

(B. Tech CSE AI -CBCS)(SUBJECTIVE TYPE)

(April, 2022)

Subject Code: BAS -107	Subject: Applied Physics-I	
Time : 3 Hours		Maximum Marks : 60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.		

Q1.		(2.5*8=20)
	(a) Check whether the electrostatic field represented by $E = a.x.y^2 (yi + xj)$ is conservative or not?	
	(b) What is the minimum number of lines per inch required in a diffraction grating to observe the second order spectra of visible light at theta =45°?	
	Data provided: Range of wavelength for Visible Light = 380 to 700 nanometers	
	(c) Calculate de Broglie wavelength of an alpha-particle whose kinetic energy is 200 eV. Given: Mass of alpha Particle = 4 a.m.u where, 1 a.m.u = 1.67×10^{-27} Kg	
	(d) Distinguish between static and dynamic response of a sensor?	
	(e) Discuss how a Four Level LASER system is better than a Three Level LASER system?	
	(f) Show that equation of continuity equation is contained in Maxwell's equations .	
	(g) If an excited state of hydrogen atom has a life-time of 2.5×10^{-14} sec, what is the minimum uncertainty with which the energy of this state can be measured? Given $h = 6.62 \times 10^{-34}$ J sec.	
	(h) Distinguish between Resistance Thermal Detector and a Thermistor?	
	UNIT-1	
Q2.	(i) Using Poynting theorem show that the power transferred into the electromagnetic field is equal to the sum of the time rate of change of electromagnetic energy within a certain volume and the time rate of the energy flowing out through the boundary surface? (ii) Calculate the value of Poynting vector at the surface of the sun if the power radiated by the sun is 3.8×10^{26} W and its radius is 7×10^{8} m. (7+3)	(10)
Q3.	(i) What are Maxwell's equations? Derive Maxwell's equations (differential form). Discuss integral form of above equations. What are the significance of these equations to electricity and magnetism? (ii) Find the total charge enclosed by a closed surface if the flux entering is 10,000 Vm and emerging out is 15,000 Vm. (7+3)	(10)
UNIT-2		
Q4.	(i) Differentiate between Fresnel and Fraunhofer's Diffraction? (ii) In Newton's rings experiment the radius of 10th and 20th rings are 0.2 and 0.3 cm, respectively, and the radius of curvature of the plano convex lens is 0.45m. Calculate the wavelength of the light used in nanometers? (6+4)	(10)
Q5.	(i) Derive Fresnel's Diffraction at a straight edge for a line source? Graphically explain why fringe width decreases with increase in order of the diffraction. (ii) Find the radius of the first half period zone of a zone plate that behaves like a convex lens of focal length 60 cm? Given $\lambda = 6000$ Å. (6+4)	(10)
	UNIT-3	
Q6.	(i) State Postulates of Quantum Mechanics? If an excited state of hydrogen atom has a life-time of 2.5×10^{-14} sec, what is the minimum uncertainty with which the energy of this state can be measured? Given $h = 6.62 \times 10^{-34}$ J sec. (ii) If light of wavelength 660 nm has wave trains 13.2×10^{-6} m long, calculate the coherence time?	(10)

Q7.	(i) For a ground state particle trapped in a 'L' dimension box, find the probability of finding the particle between x = 0 and x = L/2 dimensions. (8+2) (ii) If ground state energy of an electron trapped in one dimensional infinite potential well is 1.2 eV, calculate the energy levels of first and second excited states? (10) Given: Planck's Constant = 6.63×10^{-34} J-s and Mass of electron = 9.1×10^{-31} kg	
UNIT-4		
Q8.	What do you understand by the term "accuracy of a sensor"? List and explain the various forms in which Inaccuracies of a sensor are represented? (6+4)	(10)
Q9.	Define and Explain the following terms in context to a sensor (i) Dead Band (ii) Nonlinearity (iii) Resolution (iv) Repeatability	(10)